

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I P. Guru/192521010 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P. Guru

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.
2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.
3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy.

Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

- 1) A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.**

ANS:

Media Company – Cloud Video Streaming Platform:

Cloud Architecture:

- Deploy application on multiple cloud regions with load balancers.
- Use CDN (Content Delivery Network) to cache video content near users.
- Store videos in object storage (e.g., cloud storage buckets).
- Use auto-scaling compute instances or Kubernetes clusters.
- Store session information in distributed cache (Redis/Memcached).
- Use managed databases with read replicas.

Load Balancing and Auto-Scaling:

Load Balancer:

- Distributes user requests across multiple servers.
- Prevents server overload.
- Removes unhealthy instances automatically.

Auto-Scaling :

- Adds servers during peak traffic.
- Removes unused servers during low traffic.
- Ensures low latency and minimal buffering.
- Cost Optimization Strategies
- Use CDN caching to reduce bandwidth costs.
- Use auto-scaling to avoid paying for idle resources.
- Store older videos in low-cost archival storage.
- Use reserved/savings plans for predictable workloads.

- Compress video files using efficient codecs.
- Monitor resource usage and eliminate unused services.

Cost Optimization Strategies

To reduce cloud expenses, the company can implement several cost optimization techniques:

- Use **Auto-Scaling** to run only the required number of servers.
- Store videos in **cloud object storage** with lifecycle policies that move older content to lower-cost archival storage.
- Use a **Content Delivery Network (CDN)** to reduce bandwidth costs and improve streaming performance.
- Compress videos using efficient codecs such as **H.265 (HEVC)** or **AV1** to reduce storage and data transfer costs.
- Use **Reserved Instances** for continuously running servers and **Spot Instances** for background tasks such as video encoding and thumbnail generation.

Scalability and caching techniques:

- The system uses horizontal scaling, where additional servers are automatically added during peak traffic and removed when demand decreases.
- This ensures that the platform can handle millions of users without affecting performance.
- Caching improves response time by storing frequently accessed content in memory. Instead of retrieving the same data repeatedly from the database, the cache serves it instantly.
- A CDN also caches popular videos at edge locations worldwide, allowing users to stream content from the nearest server for faster playback.

2) A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

ANS:

Container Orchestration and Multi-Cloud Deployment Strategy:

Solution Using Container Orchestration and Multi-Cloud:

- A fintech startup deploying applications across multiple cloud providers (such as AWS, Microsoft Azure, and Google Cloud Platform) requires a consistent deployment mechanism that minimizes platform-specific differences. The most effective solution is to use Docker containers for packaging applications and Kubernetes as the container orchestration platform.
- Infrastructure can be managed using Terraform, enabling Infrastructure as Code (IaC), where identical cloud resources are provisioned across multiple providers using reusable configuration files.

For communication between microservices:

- Kubernetes DNS resolves service names.
- A service mesh such as **Istio** or **Linkerd** manages secure communication.
- Load balancing distributes requests among healthy containers.
- This eliminates manual configuration whenever new containers are deployed.

Efficient Scaling:

- Scaling is handled automatically through Kubernetes.

Horizontal Pod Autoscaler (HPA):

- Adds or removes containers depending on CPU or memory utilization.
- Maintains application performance during traffic spikes.

Cluster Autoscaler:

- Automatically adds or removes virtual machines when additional computing resources are needed.

Benefits of Multi-Cloud Deployment

- Eliminates dependency on a single cloud provider.
- Higher availability during cloud outages.
- Better disaster recovery.
- Cost optimization by selecting cheaper services.
- Compliance with regional data regulations.
- Improved performance by deploying closer to users.

Risks of Multi-Cloud Deployment

- Increased operational complexity.
- Difficult monitoring across multiple clouds.
- Security policy inconsistencies.
- Higher networking costs.
- Vendor integration challenges.
- Greater management overhead requiring skilled DevOps teams.

3) An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures

ANS:

Cloud-Native Backup, Replication, and Versioning

- An e-learning platform stores valuable information such as student records, examination data, videos, and course materials. Accidental deletion or cloud failures can lead to major service disruptions.

A cloud-native disaster recovery strategy should combine:

- Automated backups
- Cross-region replication
- Object versioning
- Disaster recovery automation

Backup Strategy:

- Automated backups are scheduled daily with incremental backups every hour.

Cloud-native services include:

- AWS Backup
- Azure Backup
- Google Cloud Backup
- Backups are stored in geographically separate regions to protect against regional failures.

Replication:

- Real-time replication copies data continuously from the primary region to a secondary region.
- Replication methods:
- Database replication
- Object storage replication
- File system replication
- If the primary region fails, users can immediately access replicated data.

Versioning:

- Versioning maintains multiple versions of files.
- Advantages:
- Recover accidentally deleted files.
- Restore previous document versions.
- Protection against ransomware.
- Easy rollback after accidental modifications.

Recovery Time Objective (RTO):

Maximum acceptable downtime.

To achieve low RTO:

Pre-configured standby servers.

Automated failover.

Infrastructure templates.

Load balancer redirection.

Example:

Target RTO = **15 minutes**

Recovery Point Objective (RPO)

Definition:

Maximum acceptable data loss.

Achieved by:

- Continuous replication
- Frequent backups
- Database transaction logging

Example:

Target RPO = **5 minutes**

Only five minutes of data may be lost.

- 4) A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.**

ANS:

Hybrid Cloud Architecture for Government Agency:

Hybrid Cloud Design:

Government agencies often handle both confidential citizen records and public information.

A hybrid cloud architecture allows:

- Sensitive applications remain on-premises.
- Public applications run in the public cloud.

Architecture:

On-Premises:

- Citizen databases
- Financial records
- Identity services
- Compliance applications.

Public Cloud:

- Public websites
- Online citizen services
- Analytics
- AI applications
- Backup storage

Both environments communicate securely through encrypted private connections.

Secure Communication:

Communication methods include:

- IPSec VPN
- Dedicated private links
- TLS encryption
- Secure API Gateways

Traffic is encrypted both during transmission and storage.

Identity Management

A centralized Identity and Access Management (IAM) solution controls authentication across both environments.

Implementation includes:

- Single Sign-On (SSO)
- Multi-Factor Authentication (MFA)
- Role-Based Access Control (RBAC)
- Least Privilege Principle
- Active Directory integration

Only authorized personnel can access sensitive systems.

Encryption:

Data at Rest:

- AES-256 encryption
- Encrypted databases

- Encrypted backups

Data in Transit:

- TLS 1.3
- HTTPS
- VPN encryption

Encryption keys are securely managed using Key Management Services (KMS).

Challenges in Hybrid Cloud:

- Complex infrastructure management.
- Network latency.
- Security policy synchronization.
- Data synchronization.
- Compliance management.
- Higher operational costs.
- Difficult monitoring across environments.
- Skilled workforce requirements.

Advantages:

- Regulatory compliance.
- Flexible workload placement.
- Improved disaster recovery.
- Scalability.
- Better resource utilization.

5) A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

ANS:

Multi-Region Cloud Deployment Strategy:

- A SaaS provider must ensure continuous service even if an entire cloud region experiences failure.
- The recommended architecture deploys application instances across multiple geographic regions

Example:

- Region 1 (Primary)
- Global Load Balancer
- Region 2 (Secondary)
- Region 3 (Disaster Recovery)

Each region contains:

- Application servers
- Databases
- Load balancers
- Storage
- Monitoring tools

Traffic is distributed to the nearest healthy region.

Data Replication

- Data must remain synchronized across regions.

Methods include:

Synchronous Replication

- Immediate updates
- No data loss
- Higher latency

Asynchronous Replication

- Faster performance
- Small possibility of data loss
- Suitable for geographically distant regions
- Databases use multi-region replication to maintain consistency.

Failover Mechanisms

Automatic failover includes:

- Continuous health checks.
- DNS failover.
- Global load balancers.
- Auto-scaling groups.
- Standby application servers.

When Region 1 becomes unavailable:

- Health check detects failure.
- Traffic is redirected.
- Secondary region becomes active.

- Users continue accessing services with minimal interruption.

Benefits:

- Nearly zero downtime.
- Fault tolerance.
- Disaster recovery.
- Global performance improvement.
- Automatic scaling.
- Improved customer satisfaction.

Challenges

- Higher infrastructure costs.
- Increased replication complexity.
- Data consistency issues.
- More complex management.
- Greater monitoring requirements.

